

AI Development Proposals

EPI Supervisor Platform

34 Innovative AI Report Templates & Features
Comprehensive Development Roadmap

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Section A: Smart Interactive Reports

1. "Field Story" - Al-Qissa Al-Maidaniyya

[High Impact - Easy Implementation]

Instead of dry tables and numbers, AI writes a narrative as if it were a field correspondent following the situation:

Today in Hudaydah, 3 directorates did not submit any report. Al-Mukalla leads with 45 reports but 12 were rejected due to missing GPS. Two children in Sayun have not received their third dose for 3 weeks.

- Merges data + geographic context + human stories
- Makes the general manager actually read the report
- Perfect for daily reports and instant notifications
- Tech: Groq LLM with data context -> narrative generation

2. "What If?" Simulator - Al-Muhakat

[Strategic Impact - Medium Implementation]

AI calculates the impact of different scenarios before making decisions:

If we raise Al-Mahra coverage from 76% to 90% -> we need 340 additional doses, 8 new sessions, estimated cost \$2,400, and 2 extra teams for 1 week.

- Supports decisions: "Add a team?", "Expand campaign?", "Change timing?"
- Calculates required doses, resources, cost, and time

3. "Voice Report" - Al-Taqrir Al-Sawti

[User Experience - Easy Implementation]

Instead of reading -> AI summarizes the report as audio:

- Manager listens while moving between centers
- 60-second summary: submissions, critical items, recommendation
- Supports Modern Standard Arabic and Yemeni dialect
- Tech: TTS on LLM output

4. "Data Conversation" - Muhadathat ma' Al-Bayyanat

[Interactive Experience - Medium Implementation]

User converses directly with the data:

User: "How many children in Hudaydah didn't get Penta3?"
 AI: "1,086 children. 67% are in two directorates: Bait Al-Faqih and Al-Munira."
 User: "What's the reason?"
 AI: "Analysis shows: 52% distance, 31% displacement, 17% parental refusal."

- Interactive chat with database - no SQL needed
- Supports follow-up questions with contextual memory

Section B: Narrative & Story Reports

5. "Dropout Forensics" - Tahliy AI-Insihab

[Direct Coverage Impact - Medium Implementation]

AI analyzes every lost child and classifies reasons:

1,247 children started Penta1 but didn't complete Penta3:
 45% in Al-Mukalla -> Reason: distance (3 centers too far without transport)
 30% in Hudaydah -> Reason: family displacement last month
 25% varied -> Individual reasons

- Discovers dropout patterns: geographic, seasonal, demographic
- Suggests targeted interventions for each reason

6. "Supervisor Scorecard" - Bitaqat Ada' Al-Mushrif

[HR Management - Easy Implementation]

AI-generated comprehensive performance card for each supervisor:

Ahmed Mohammed - Hudaydah Directorate:
 34 submissions this month (team avg: 21)
 Approval rate: 94% (team: 78%)
 Avg input time: 4.2 min
 100% GPS-enabled submissions
 AI: Candidate for promotion or peer training

- Monthly scorecard for each supervisor and area manager
- Compares with team average and previous month

7. "Executive Brief" - Waraqat Al-Mukhtasar Al-Tanfidhi

[For Senior Leadership - Easy Implementation]

One page only - designed for officials with no time:

- Overall health: one point (green/yellow/red)
- Only 3 key numbers
- Biggest challenge + biggest achievement
- One actionable recommendation
- Designed to be read in under 60 seconds

Section C: Predictive & Simulation Reports

8. "Risk Heatmap" - Kharitat Al-Khatar

[Proactive - Medium Implementation]

AI classifies governorates/directorates by risk level:

RED: Hudaydah - coverage 62% + high population density + no campaign in 8 months

YELLOW: Mahra - coverage 78% but rising dropout

GREEN: Abyan - coverage 94% + recent successful campaign

- Combines: coverage + population density + campaign dates + epidemiology
- Identifies where to focus resources immediately

9. "Outbreak Early Warning" - Indhar Tafashshy Mubakkir

[Critical Public Health - Medium Implementation]

AI warns 7-14 days before potential outbreak:

WARNING: Hudaydah - MR coverage dropped from 76% to 71% in 3 weeks + 3 suspected measles cases. If decline continues -> outbreak risk within 10 days. Recommendation: Immediate preventive MR campaign.

- Monitors: coverage drop + suspected cases + seasonality + displacement
- Classifies: Warning / Alert / Emergency

10. "Goal Trajectory" - Taqaddum Nahwa Al-Hadaf

[Strategic Planning - Easy Implementation]

AI calculates whether we will reach the target and when:

Annual target: Penta3 coverage = 95%

Current: 90% (month 4 of 12)

Curve: Rising at 1.1%/month

Projection: 93.3% year-end (1.7% below target)

Needed: Accelerate by 1.5%/month = ~800 extra doses/month

11. "Optimal Timing" - Al-Tawqit Al-Mithali

[Operations Optimization - Easy Implementation]

AI discovers best input times from historical data:

6-month analysis:

Best day: Sunday (avg 45 submissions)

Best time: 9-11 AM (quality 23% higher)

Worst time: After 3 PM (rejection 31% higher)

Suggestion: Focus sessions Sun-Thu mornings

Section D: Geographic & Risk Reports

12. "District Coverage Map" - Kharitat Takatthul Al-Mudiriyyat

[Visualization - Medium Implementation]

Interactive heatmap for each district with accompanying report:

- Colors: Green (>90%) / Yellow (70-90%) / Red (<70%)
- Click district -> AI writes its analysis
- Compare with neighboring district
- Suggest centers needing visits

13. "Optimal Route" - Masar Al-Jawla Al-Mithali

[For Field Supervisors - Medium Implementation]

AI suggests the best order for field visits:

- Optimize geographic route (TSP solver)
- Prioritize low-coverage centers
- Consider capacity of each center
- Dynamic rerouting based on new data
- Saves time and fuel - very useful in rural areas

14. "Coverage Gap Analysis" - Tahliy Al-Fajat

[Strategic Analysis - Easy Implementation]

AI identifies unserved areas:

- Directorates without nearby vaccination center
- High density areas with low coverage
- Displacement lines affecting coverage
- Suggest new center locations

Section E: Individual Performance Reports

15. "Data Entry Leaderboard" - Lawhat Sharaf Al-Mudkhilin

[Motivation - Easy Implementation]

Monthly ranking of data entrants by performance:

- Top 3 with medals
- Points: submissions x approval rate x GPS completeness
- Achievement badges: "Star of Week", "Highest Quality"
- Public board in app - creates healthy competition

16. "Personal SWOT" - Quwwat wa Da'f Shakhsi

[Professional Development - Medium Implementation]

AI analyzes each entrant's performance and writes SWOT:

- Strengths: input speed, field completeness
- Weaknesses: GPS errors, submission delays
- Opportunities: training on new forms
- Threats: burnout, campaign pressure

17. "Activity Tracker" - Raqib Al-Nashat

[Management - Easy Implementation]

Track each user's activity:

- Last login + session duration
- Submissions per day
- Compare with previous week/month
- Auto-alert if inactive >3 days

Section F: Motivation & Competition Reports

18. "Governorate Championship" - Abtalat Al-Muhafazat

[Group Motivation - Easy Implementation]

Sports-style competitive report that motivates teams:

Gold: Abyan - Champion this week (+12% improvement)

Rising star: Lahj (+18% submissions)

Biggest decline: Mahra (-22%)

Next challenge: Can Hudaydah break 80%?

19. "Vaccination Race" - Sabaq Al-Tat'im

[Gamification - Easy Implementation]

Turns vaccination numbers into a fun race:

- Each governorate "runs" toward the goal
- Live progress bars
- Notifications: "Lahj overtook Mahra!"
- Celebration at 100%

Section G: Quality & Analysis Reports

20. "Rejection Autopsy" - Tashrih Al-Rafd

[Quality Improvement - Medium Implementation]

AI analyzes reasons for every rejected submission:

Analysis of 340 rejected submissions:

41% missing GPS -> 3 directorates responsible for 70%

28% illogical dates (future dates! -> device settings)

19% duplicate photos -> duplicate entry

12% incomplete child names

21. "Fraud Detection" - Kashf Al-Tazwir

[Security - Medium Implementation]

AI detects suspicious submissions:

- Duplicate or edited photos (image analysis)
- Repeated or geographically impossible GPS
- Too-fast entries (under 30 seconds)
- Unusual input patterns
- Classifies: Warning / Review / Temporary block

22. "Data Completeness" - Ikmal Al-Bayyanat

[Quality - Easy Implementation]

Measures completeness of each field in each form:

- GPS, photos, notes completion rates
- Which directorates leave fields empty
- Did completeness improve after awareness campaign?

Section H: Epidemiological & Health Reports

23. "Epidemiological Pulse" - Nabdh Waba'i

[Public Health - Medium Implementation]

Combines platform data with epidemiological indicators:

Measles: 3 suspected in Hudaydah (Trend UP)
 Polio: zero cases - excellent performance
 TB: 2 AEFI notices (normal - local reaction)
 Recommendation: Preemptive MR campaign in Hudaydah

24. "AEFI Report" - Taqir Hawadith Ma'ba'd Al-Tat'im

[Safety - Easy Implementation]

Analysis of adverse events following immunization:

- Classify by type and severity
- Patterns linked to specific vaccines?
- Compare with global rates
- Safety recommendations

25. "Cold Chain Report" - Taqir Silsilat Al-Barid

[Logistics - Easy Implementation]

Monitor vaccine storage and transport:

- Recorded temps vs recommended (2-8C)
- Alerts when limits exceeded
- Current vaccine stock per center
- Predict depletion based on consumption rate

Section I: Timing & Planning Reports

26. "Campaign Lifecycle" - Dawrat Hayat Al-Hamla

[Campaign Management - Medium Implementation]

AI monitors supplementary campaign in real-time:

Polio Campaign - Round 3:
 Day 12 of 21 -> 57% complete
 8,432 children of 14,800 targeted

Warning: 3 directorates below 40% in first half

AI Suggestion: Redirect 2 teams from completed to slow areas

27. "Smart Session Planner" - Mukhatit AI-Jalsat AI-Dhaki

[Planning - Medium Implementation]

AI suggests optimal schedule for upcoming sessions:

- Which centers need extra sessions
- Best days and times for each directorate
- Team allocation across centers
- Account for holidays and official breaks

Section J: Regional & International Reports

28. "Regional Benchmark" - Muqarana Iqlimiyya

[For International Reports - Easy Implementation]

Comparison with WHO regional standards:

Yemen vs Region:

OK Penta3: 90% (Regional avg: 82%) - Above average!

WARN MR1: 86% (Regional avg: 89%) - Slightly below

RED Dropout: 6.3% (Regional best: 2.1%) - Improvement opportunity

- Report ready for donors and international bodies
- Integrated WHO/UNICEF data

29. "Donor Progress Report" - Taqrir Taqaddum Lil-Mawmin

[Donor Relations - Easy Implementation]

- Achievement summary against agreed targets
- Field success stories (from AI Field Story)
- Funds used vs results
- Plan for next quarter
- Professional PDF ready to send

Section K: Mobile App Features

30. "Smart Form Fill" - Taml' Dhaki

[UX - Medium Implementation]

- AI auto-suggests field completions
- When entering child name -> suggests from previous data
- Auto GPS + location verification
- Alert on illogical data (age 200 years!)

31. "Smart Notifications" - Ilam Dhaki

[Notifications - Easy Implementation]

- No spam - AI decides when to notify
- Priority: Critical immediate / Important today / Informational
- Smart timing: no notifications after 9 PM except critical
- Daily digest instead of 20 separate alerts

32. "Voice Input" - Idkhal Sawti

[Accessibility - Medium Implementation]

- Field supervisor speaks -> AI fills fields
- Useful in rural areas with difficult writing
- Supports Arabic and local dialect
- Tech: Whisper API or Groq Whisper

33. "Auto Translation" - Tarjama Awtumatiq

[i18n - Easy Implementation]

- Reports auto-translate to English
- For international reports and foreign donors
- Preserves correct technical terminology

34. "Field Copilot" - Musa'id Al-Maidani

[For Mobile App - Medium Implementation]

- Built-in chat in mobile app
- Ask: "Is this center complete?", "How many left today?"
- Works fully offline
- Smart suggestions during data entry

Section L: Development Roadmap

Phase 1: Quick Wins (2 Weeks)

[Immediate Impact]

- #1 Field Story
- #6 Supervisor Scorecard
- #7 Executive Brief
- #15 Data Entry Leaderboard
- #18 Governorate Championship
- #22 Data Completeness Report

Phase 2: Core Intelligence (1 Month)

[Foundation]

- #4 Data Conversation
- #5 Dropout Forensics
- #8 Risk Heatmap
- #9 Outbreak Early Warning
- #20 Rejection Autopsy
- #26 Campaign Lifecycle

Phase 3: Advanced Features (2 Months)

[Advanced]

- #2 What-If Simulator
- #10 Goal Trajectory Predictor
- #12 Interactive Coverage Map
- #13 Optimal Route Planner
- #21 Fraud Detection
- #32 Voice Input

Phase 4: Ecosystem (3 Months)

[Full System]

- #3 Voice Report (TTS)
- #28 Regional Benchmark
- #29 Donor Progress Report

- #30 Smart Form Fill
- #31 Smart Notifications
- #34 Field Copilot